

SAFETY DATA SHEET (GHS)

Nitrogen (compressed / refrigerated liquid) | CAS: 7727-37-9 | Formula: N₂

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Nitrogen, CAS 7727-37-9 (<https://echa.europa.eu/>)
- PubChem CID 947 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/947>
- ILO/NIOSH International Chemical Safety Card (ICSC) 0110 — Nitrogen
- NOAA CAMEO Chemicals — Nitrogen datasheet (<https://cameochemicals.noaa.gov/>)
- Supplier SDS — Nitrogen, compressed and refrigerated liquid grades, multiple suppliers
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Nitrogen (compressed gas and/or refrigerated liquid)

CAS Number: 7727-37-9

Chemical formula: N₂

Molecular weight: 28.01 g/mol

Synonyms: Liquid nitrogen (LIN); GAN (gaseous nitrogen)

Intended use: Inert gas blanketing/purging for storage tanks, reactors and process vessels (notably critical for CHP storage tank D-2301 series and other peroxide-containing equipment) and general utility inert gas supply. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Press. Gas — Compressed gas or Refrigerated liquefied gas (H280 / H281, depending on supply form)

Signal word: Warning

Hazard statements (H-codes): H280 (Contains gas under pressure; may explode if heated) — for compressed gas cylinders/pipework; H281 (Contains refrigerated gas; may cause cryogenic burns or injury) — for liquid nitrogen (LIN) supply

Precautionary statements (P-codes): P282, P336, P403, P410+P403

GHS pictograms: GHS04 (Gas cylinder)

Other hazards: PRIMARY HAZARD IS ASPHYXIATION — nitrogen is chemically inert but displaces oxygen; an atmosphere of essentially pure nitrogen provides NO warning properties (odourless, colourless, tasteless) and can cause rapid loss of consciousness and death within seconds to minutes upon entry to an oxygen-deficient space (e.g. inerted/purged vessels, confined spaces adjacent to nitrogen vents). This hazard is especially significant given the widespread use of nitrogen blanketing across Train II — vessel entry procedures and confined space controls are critical. Liquid nitrogen (-196°C) causes severe cryogenic burns/frostbite on contact and can embrittle ordinary carbon steel and many other materials not rated for cryogenic service. Rapid vaporisation of LIN (expansion ratio ~1:694) in a confined space can also generate significant overpressure.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Nitrogen: CAS 7727-37-9 — typically >99.9% (industrial grade); confirm purity specification for inerting/blanketing applications vs. any instrumentation-grade requirements

SECTION 4 — FIRST AID MEASURES

Inhalation: CRITICAL — if a person collapses in a nitrogen-inerted atmosphere, DO NOT enter without appropriate respiratory protection (SCBA) and a trained rescue team — multiple fatalities have occurred from would-be rescuers succumbing to the same oxygen-deficient atmosphere. Remove the victim to fresh air using proper rescue equipment, administer oxygen and CPR/artificial respiration as needed, and obtain immediate medical attention.

Skin contact: For contact with liquid nitrogen or cold-gas-contacted surfaces (cryogenic burn/frostbite): flush affected area with lukewarm (not hot) water; do not rub the affected area; remove non-adhered clothing; obtain medical attention for any significant cold burn

Eye contact: For cryogenic splash/frostbite: flush gently with lukewarm water for at least 15 minutes; obtain medical attention

Ingestion: Not applicable (gas/cryogenic liquid)

Notes to physician: Treat as hypoxic/anoxic injury (assess neurological status, cardiac rhythm) for inhalation incidents; treat cryogenic contact as a cold-thermal burn

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Not combustible — nitrogen itself does not burn and is in fact used as a fire-suppression/inerting medium; use extinguishing media appropriate to the surrounding fire

Unsuitable extinguishing media: Not applicable

Special hazards: Non-flammable, but cylinders/pipework/storage vessels exposed to fire can rupture due to pressure buildup (BLEVE risk for liquid nitrogen vessels if relief devices are inadequate/blocked). Large LIN releases displace oxygen over a wide area and can obscure visibility (dense white vapour cloud from condensed atmospheric moisture).

Special protective equipment for firefighters: Standard SCBA; be aware that a fire in an area with significant nitrogen release may itself create an oxygen-deficient atmosphere even outside the immediate fire zone

Hazardous combustion products: None — nitrogen is inert and non-combustible

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Evacuate the area. Ventilate to restore normal oxygen levels before entry — use portable oxygen monitors to confirm safe atmosphere (>19.5% O₂) before entry without SCBA. Treat any enclosed or low-lying area near a nitrogen release as a potential oxygen-deficient atmosphere until proven otherwise.

Environmental precautions: None — nitrogen is a normal atmospheric constituent (78% of air) and poses no environmental hazard

Containment and cleanup: Allow gas to disperse/vent to a well-ventilated area; for LIN spills, keep personnel away until the cryogenic liquid has fully vaporised and the area has been confirmed safe (oxygen level and temperature)

Reference to other sections: See Section 7 (Handling/Storage) and Section 8 (confined space/oxygen monitoring controls)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Ensure adequate ventilation in all areas where nitrogen is used for blanketing/purging, especially confined spaces, pits, and below-grade areas where nitrogen (denser than air when cold, similar density to air at ambient temperature) can accumulate. For liquid nitrogen: use only equipment rated for cryogenic service; allow for thermal contraction/expansion in piping design.

Storage conditions: Compressed gas cylinders/pipework: store/operate within rated pressure and temperature limits, secured against falling, away from heat sources. Liquid nitrogen (LIN) storage: vacuum-insulated cryogenic vessels with appropriately sized pressure relief devices (PRDs) vented to a safe, well-ventilated location — NEVER fully close off a LIN system without functioning relief, due to the large expansion ratio on vaporisation.

Temperature limits: LIN at approx. -196°C; ensure all wetted materials (piping, valves, gaskets) are rated for cryogenic service to avoid brittle fracture

Pressure: Per system design — compressed gas systems typically operate at moderate pressure for blanketing/purging applications; confirm relief device sizing for both compressed gas and LIN systems

Segregation requirements: No chemical segregation requirements (inert gas); physical segregation/labelling of nitrogen-purged equipment from air/oxygen systems is important to prevent inadvertent inerting hazards to personnel

Special requirements: Confined space entry procedures with mandatory oxygen monitoring (and LEL/toxic gas monitoring as applicable) for ANY vessel that has been or may be nitrogen-purged/blanketed — this applies across the Train II inventory wherever inert gas blanketing is used (notably CHP storage and other peroxide-containing equipment). Positive isolation (double block and bleed, line blinding) of nitrogen supply to vessels before personnel entry.

Specific end-use recommendations: For CHP storage tank blanketing specifically, ensure the nitrogen blanketing system has adequate redundancy/alarm (pressure/flow) given the criticality of maintaining an inert atmosphere over the peroxide inventory — loss of blanket gas combined with air ingress is a recognised contributory hazard for organic peroxide storage

(see CHP SDS, CAS 80-15-9)

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): Not applicable — nitrogen is a normal atmospheric constituent with no toxicity-based OEL; managed via oxygen concentration monitoring (minimum acceptable oxygen level typically 19.5% by volume per most confined space standards)

TWA: Not applicable

STEL: Not applicable

Ceiling: Not applicable

Skin notation: Not applicable

IARC/NTP/OSHA carcinogen status: Not applicable — not a carcinogen

Engineering controls: Continuous oxygen monitoring in confined spaces and any area where nitrogen accumulation is possible; adequate ventilation; interlocked/alarmed nitrogen blanketing systems on critical vessels

PPE — Respiratory: SCBA or supplied-air respirator required for entry into any atmosphere with oxygen <19.5% (or per local confined space standard); not required for normal handling of properly contained nitrogen systems

PPE — Gloves: Insulated/cryogenic-rated gloves for any task involving contact with liquid nitrogen or cold-contacted equipment

PPE — Eyes: Face shield/goggles for tasks with potential for LIN splash (cryogenic burn protection)

PPE — Body: Standard work clothing for gas handling; insulated/cryogenic apron and footwear for LIN transfer operations

Biological monitoring: Not applicable

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless gas; colourless cryogenic liquid (LIN)

Odour: Odourless — NO WARNING PROPERTIES for oxygen-deficiency hazard

Odour threshold: Not applicable (odourless)

pH: Not applicable

Melting point / Freezing point: -210.0°C

Boiling point (at 1 atm): -195.8°C

Flash point: Not applicable — non-flammable

Evaporation rate: Not applicable (gas); LIN vaporises rapidly at ambient temperature with a liquid-to-gas expansion ratio of approximately 1:694

Flammability (solid/gas): Non-flammable; will not support combustion (used as an inerting medium)

Flammability limits in air: Not applicable (non-flammable)

Vapour pressure (at 20°C): Not applicable in the relevant sense for a permanent gas at ambient temperature; LIN vapour pressure follows its own saturation curve below -195.8°C

Vapour density (air = 1): ~0.97 (slightly lighter than air at ambient temperature; cold nitrogen vapour from LIN releases is initially denser than ambient air and can pool at grade before warming and dispersing)

Relative density (water = 1): ~0.81 (liquid nitrogen, at boiling point)

Solubility (water): ~18 mg/L at 25°C (slightly soluble; not relevant to hazard profile)

Partition coefficient (log Kow): Not applicable (elemental diatomic gas)

Auto-ignition temperature: Not applicable (non-flammable)

Decomposition temperature: Not applicable (stable diatomic molecule)

Viscosity: Not applicable (gas); LIN viscosity very low

Explosive properties: Not classified as explosive; pressurised/cryogenic vessel rupture (physical, not chemical, explosion) is the relevant energy-release hazard

Oxidising properties: Not an oxidiser — nitrogen is inert and is used specifically to EXCLUDE oxidisers (oxygen) from process equipment

Molecular weight: 28.01 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Chemically inert under normal conditions; nitrogen is one of the least reactive common gases and does not participate in combustion or most chemical reactions at ambient temperature

Chemical stability: Stable

Possibility of hazardous reactions: None under normal conditions. At very high temperatures (e.g. in combustion processes or electrical arcs), nitrogen can react with oxygen to form nitrogen oxides (NO_x) — not relevant to normal inert-gas blanketing service

Conditions to avoid: None from a chemical reactivity standpoint; from a physical standpoint, avoid uncontrolled release into confined/enclosed spaces (asphyxiation) and avoid exposing non-cryogenic-rated equipment to liquid nitrogen temperatures (brittle fracture)

Incompatible materials: None chemically; materials not rated for cryogenic temperatures (e.g. ordinary carbon steel, some plastics/elastomers) are 'incompatible' with liquid nitrogen in the sense of becoming brittle and failing

Hazardous decomposition products: None

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation (asphyxiation via oxygen displacement — the only significant toxicological route); skin/eye contact with cryogenic liquid (physical cold-burn injury, not chemical toxicity)

Acute toxicity — Oral LD50: Not applicable

Acute toxicity — Dermal LD50: Not applicable

Acute toxicity — Inhalation LC50: Not applicable — nitrogen itself is non-toxic; the hazard is purely the displacement of oxygen. Effects of oxygen deficiency: ~12-16% O₂ — increased breathing/heart rate, impaired coordination; ~10-14% O₂ — impaired judgement, rapid fatigue; ~6-10% O₂ — loss of consciousness within minutes; <6% O₂ — loss of consciousness almost immediately, death within minutes

Skin corrosion/irritation: Not classified (chemical); cryogenic contact causes physical cold-burn/frostbite injury

Serious eye damage/irritation: Not classified (chemical); cryogenic splash causes physical cold-burn injury

Respiratory/skin sensitisation: Not applicable

Germ cell mutagenicity: Not applicable

Carcinogenicity: Not applicable — not a carcinogen

Reproductive toxicity: Not applicable

STOT (single exposure): Not classified (chemical) — see asphyxiation effects above, which are physiological/physical, not a chemical STOT effect

STOT (repeated exposure): Not applicable

Aspiration hazard: Not applicable

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Not applicable/not classified — nitrogen is a natural atmospheric constituent and is non-toxic to aquatic organisms at relevant concentrations

Persistence and degradability: Not applicable (elemental gas, atmospheric constituent)

Bioaccumulative potential: Not applicable

Mobility in soil: Not applicable (gas, disperses to atmosphere)

Other adverse effects: None — no ozone depletion potential, not a greenhouse gas of concern in this context

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Release excess gas to atmosphere in a well-ventilated outdoor location away from air intakes and occupied areas, in accordance with facility venting procedures; no special waste disposal required (inert, non-hazardous)

Contaminated packaging: Not applicable (gas)

Regulatory reference: Not regulated as a hazardous substance for disposal purposes

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1066 (Nitrogen, compressed) or UN 1977 (Nitrogen, refrigerated liquid) — confirm per supply form

Proper shipping name: NITROGEN, COMPRESSED or NITROGEN, REFRIGERATED LIQUID

Transport hazard class(es): Class 2.2 (Non-flammable, non-toxic gas)

Packing group: Not applicable (gases)

Environmental hazards (marine pollutant): Not classified as a marine pollutant

Special precautions for user: Ensure cryogenic vessels (LIN) have functioning pressure relief devices at all times during transport; secure cylinders against falling

Bulk transport (MARPOL/IBC): Classify per IGC Code if shipped in bulk by sea as refrigerated liquefied gas

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below. Confined space entry regulations (oxygen monitoring requirements) are the primary regulatory driver for nitrogen handling at this facility.

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 7727-37-9)

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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